

Cognex Vision System Basics

Lecture 2

Most Recognized Machine Vision Companies

- Keyence Corporation (Japan)
- Teledyne Technologies (USA)
- Basler (Germany)
- Omron Corporation (Japan)



Who is **COGNEX**?

- **Cognex Corporation** develops machine vision systems, vision sensors, and industrial barcode readers used in automated manufacturing and logistics.



Who is **COGNEX**?

- Founded in 1981 and based in Natick, Massachusetts, the company provides tools for applications such as inspection, identification, and guidance.
- Its products are used across various industries to support quality control and process automation.
- Cognex also offers software platforms like In-Sight Explorer and VisionPro for configuring and deploying vision-based solutions.

Cognex Machine Vision Products



2000 Series



8400 Series



7000 Series



9000 Series

Machine Vision

Vision Sensors

In-Sight SnAPP
In-Sight 2000

Vision Systems

In-Sight 2800
In-Sight 3800
In-Sight 3800 Line Scan
In-Sight 8900
In-Sight 7000
In-Sight 8000
In-Sight 9000

3D Vision Systems

In-Sight L38
L38 with VisionPro
L68
3D-A5000

OEM Products

Advantage 182
Advantage 100
Advantage Vision Engine

Cognex Camera

- Depending on the model, an In-Sight device is technically called a “sensor” or “vision system,” but we often refer to it informally as a “**camera**.”
- You program an In-Sight camera via an application called **In-Sight Explorer**, which runs on a PC that is networked to the camera.
- The programming ‘language’ is actually a **spreadsheet**, very similar to Excel. We call the spreadsheet and its contents a **job**.



In-Sight Explorer

- In-Sight Explorer has two modes:
 - **Spreadsheet Mode**, in which you set up the spreadsheet itself.
 - **EasyBuilder Mode**, in which you go through a series of menu steps that create the spreadsheet for you.

In-Sight Explorer

- Once the job has been created, you do not need to have a PC on the network to run the job.
- You can create, edit, and view jobs from In-Sight Explorer. More than one camera can be viewed at time by means of In-Sight Explorer's tiling ability in its **Windows** pull-down menu.
- A camera can only be connected to a single In-Sight Explorer instance.

Choices in In-Sight Camera Features

- Processor speed (Simple or Complicated task)
- HDR (High Dynamic Range) and HDR+ to enhance contrast (More Realistic images)
- Resolution: 640x480 pixels - 4000x3000 pixels
- Vision Tools Included: all tools or ID tools only
- Number of Discrete Inputs and Outputs
- Image: Greyscale or Color; Area Scan or Line Scan
- Integrated lighting
- Auto-focus lens
- PoE (Power over Ethernet)
- Physical size: Micro (30 x 30 x 60mm) and up

I/O Expansion Modules

- Each model of In-Sight has a built-in number of discrete I/O lines.
- More lines can be made available on most models by adding an I/O Expansion Module. It extends the capabilities by adding discrete inputs/outputs, and hardware handshaking for serial communications.



I/O Expansion Modules

- The CIO-MICRO module includes
 - 1 trigger input
 - ON: 20 to 28V (24V nominal), <7.5 mA
 - OFF: 0 to 3V (8V nominal threshold), <250 μ A; Resistance ~10,000 Ohms
 - 8 discrete inputs (Maximum 30 VDC, 104 mA)
 - 8 discrete outputs (Maximum 30 VDC, 100 mA)



In-Sight Explorer in EasyBuilder Mode

- The **EasyBuilder Interface** allows you to go through a series of menu steps that create the spreadsheet for you.

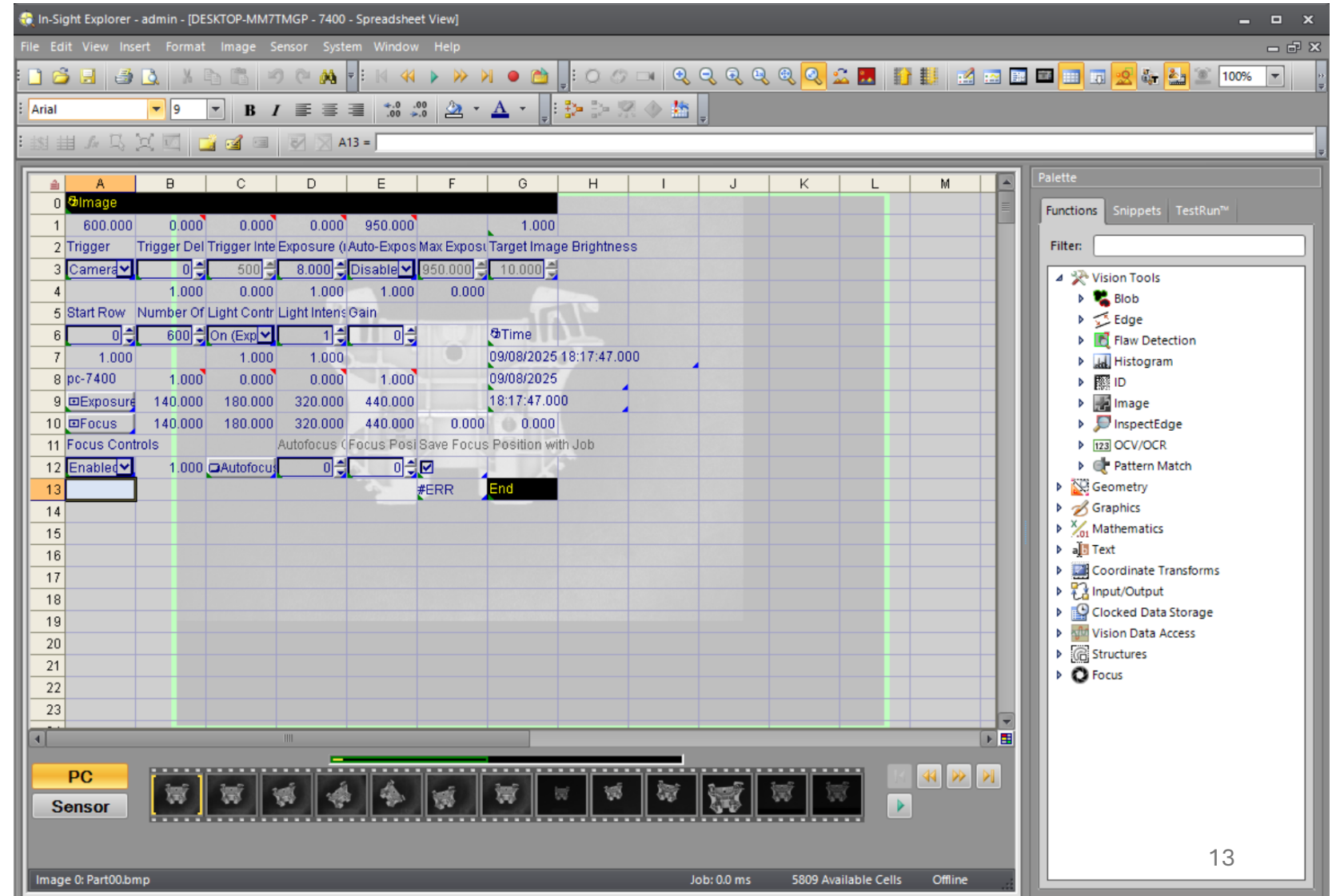
The screenshot displays the In-Sight Explorer software interface in EasyBuilder Mode. The main window shows a camera feed of a Cognex camera with a green bounding box around the text 'COGNEX' and 'ABC123'. A large green checkmark and the word 'PASS' are visible on the screen. The interface includes a menu bar (File, Edit, View, Image, Sensor, System, Window, Help), a toolbar, and a sidebar with 'Application Steps' (1. Start, 2. Set Up Tools, 3. Configure Results, 4. Finish) and a 'Tool' list (Presence/Absence Tools, Measurement Tools, Counting Tools, Identification Tools, Geometry Tools, Math & Logic Tools, Plot Tools, Image Filter Tools, Defect Detection Tools, Calibration Tools). The 'Results' tab is active, showing a table of results:

Name	Result	Type
Locator	(67.1,24.8) 0.0° score = 100.0	PatMax® Pattern
Connector	52.902	Brightness
Edge_1_Wid...	Present	Edge
Edge_2_Wid...	Present	Edge
Width_1	44.251 mm	Distance
Count_LEDs	3.000	Blobs
IDCode_1	ABC123	Read 2D Code
Binimize	Pass	Point

The bottom right corner shows 'NA' and 'Time: 0.0ms'. The bottom left corner shows '72% Job Size Available' and 'Offline'.

In-Sight Explorer in Spreadsheet Mode

- The **Spreadsheet Interface** allows you to set up the spreadsheet itself. This allows for customization to fit your specific needs.

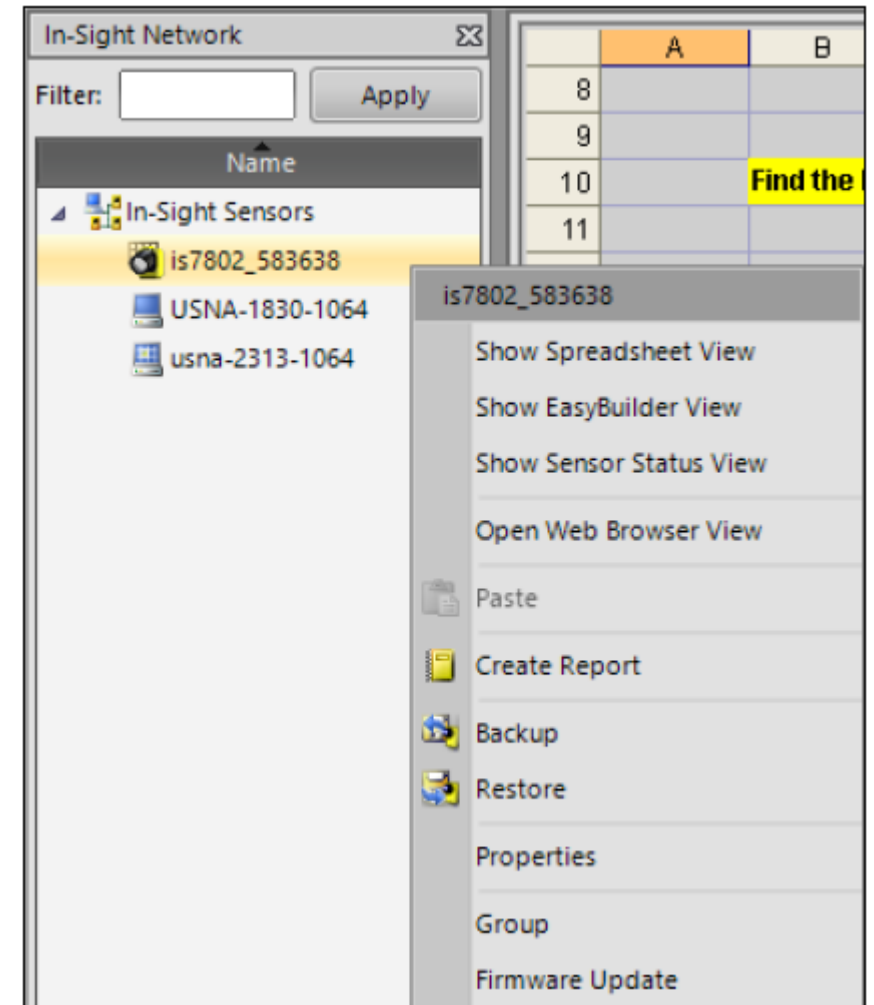


EasyBuilder Mode vs. Spreadsheet Mode

- The benefits of the EasyBuilder Interface include:
 - Provides intuitive setups for applications.
 - You do not see the spreadsheet. EasyBuilder creates it for you.
 - Makes even the most powerful vision tools easy to use.
- The benefits of the Spreadsheet Interface include:
 - Allows full access to all In-Sight functionality.
 - Allows creation of custom graphical user interface (Custom View).
 - Allows complex logic statements.
 - Can be slightly faster.
 - Scripting capabilities (JavaScript).

Switching Modes

- In-Sight Explorer opens EasyBuilder view by default.
- Right-clicking an In-Sight device from the In-Sight Network panel displays a menu that allows switching between Spreadsheet and EasyBuilder views.
- It can also be done through Window Tap.



TCP/IP and IP Address

- **TCP/IP** – Transmission Control Protocol / Internet Protocol, is a set of communication protocols that allows devices to communicate with each other over a network.
- **IP Address** – Internet Protocol address, is a numerical address assigned to a device/network interface that allows it to be identified and reached on an IP network. A typical IP address has the form **xxx.xxx.xxx.xxx**, where each **xxx** is a number from **0 to 255**. Example: 192.168.0.5.
- **NOTE:** *Microsoft's default TCP/IP network is 192.168.0.XXX.*

Static IP Address

- **IP address and subnet mask are manually configured** by a person (e.g., Network Administrator)
- **Usually stays the same** after power cycling or restarting
- **Requires manual configuration** for each device
- Easier to **reliably access the device at a known IP address**
- Useful for devices that need a **consistent network address**
- **More administrative effort** when many devices are involved
- Risk of **IP address conflicts** if addresses are assigned incorrectly

Dynamic IP Address

- **IP address and subnet mask are automatically assigned**, typically by a **DHCP server** (Dynamic Host Configuration Protocol, is a network protocol that automatically assigns IP addresses and other network configuration parameters to devices connected to a network.)
- IP address **may change** after a restart, power cycle, or when the DHCP lease expires
- **Little or no manual configuration** is required
- Easier to manage when there are **many devices**
- Generally **reduces the chance of IP address conflicts** because the DHCP server manages address assignments
- A device's IP address is **not necessarily predictable**

In-Sight Camera IP Addressing

- In-Sight cameras can be configured to use either **Static or Dynamic IP addressing**.
- A new or repaired camera is shipped with **DHCP enabled** by default.
- **When should we use a Static IP address?**
- Imagine a PLC is programmed to communicate with a camera at IP address **192.168.1.100**.
- If the camera uses DHCP, its IP address may change to **192.168.1.135**.
- The PLC may then no longer be able to communicate with the camera because it is still trying to connect to **192.168.1.100**.
- Therefore, a **Static IP address is often preferred for industrial devices** that need to be reliably accessed by other devices.

Subnet

- The **Subnet** is a group of networked In-Sight cameras with similar IP addresses.
- **Subnet Mask** defines which part of the IP address refers to the network and which part refers to the host (Camera, PC, Robot, PLC, etc.).
- The subnet mask must be the same for all devices on a network. Example: 255.255.255.0.
- **NOTE:** *255 means all IP addresses on this subnet are identical in this position. 0 means each IP address is different in this position.*
- In a static network, a person specifies the subnet mask. In a dynamic network, the DHCP server assigns the subnet mask.

Subnet

- There are three classes of subnet masks. The class of a particular subnet on a network is defined by the number of bits used to represent the network and host address portions in the IP address, as in the table below:

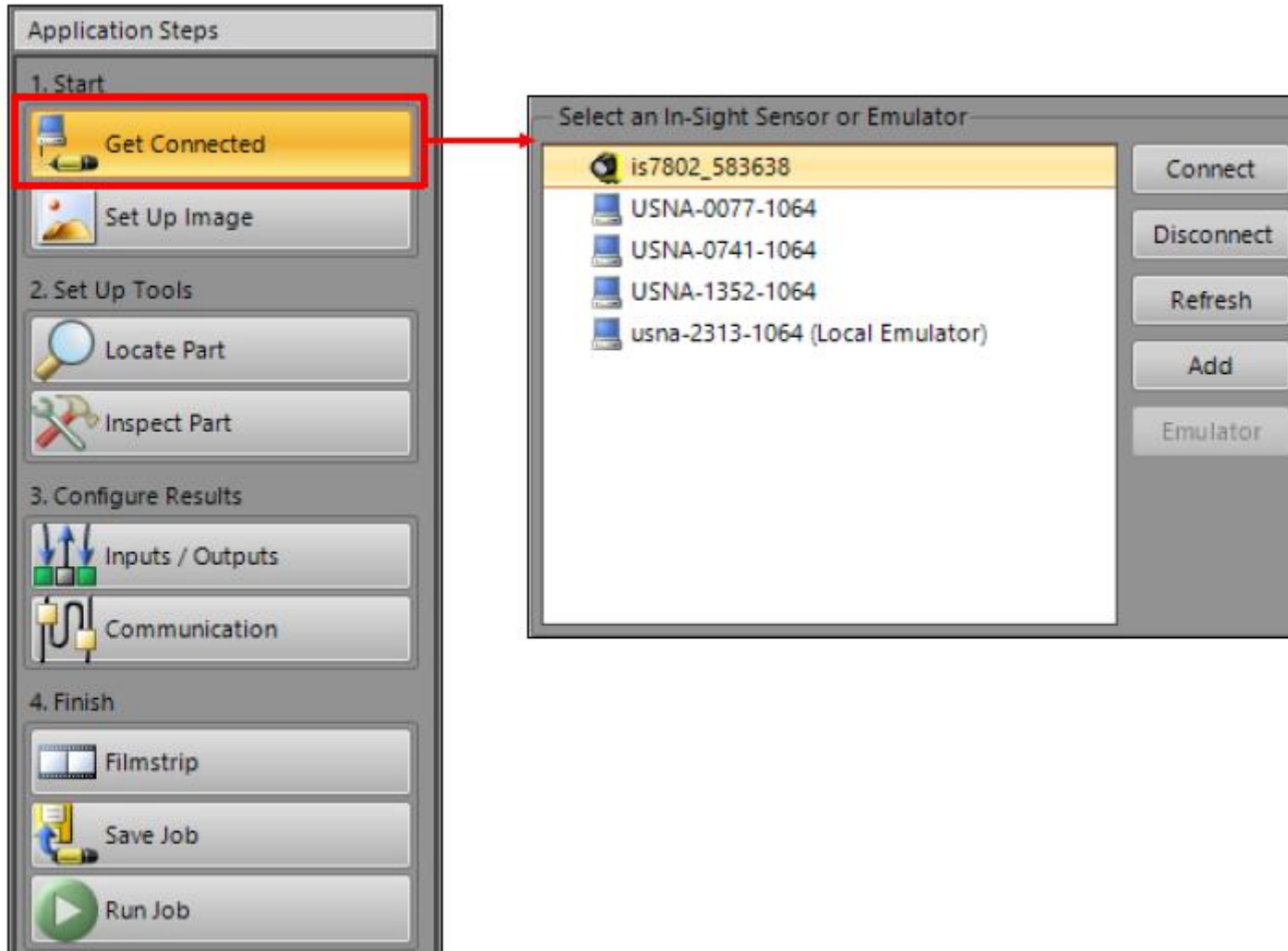
Class	Subnet Mask	Network Address	Host Address
A	255.0.0.0	8 bit	24 bit
B	255.255.0.0	16 bit	16 bit
C	255.255.255.0	24 bit	8 bit

Subnet

- For example, consider a networked In-Sight host system with the IP address 192.168.0.1. If the first three numbers (192.168.0) identify the 24 bit network address, and the last number (1) is the 8 bit address for the In-Sight host on the network, then the subnet mask for this host is 255.255.255.0 (Class C).

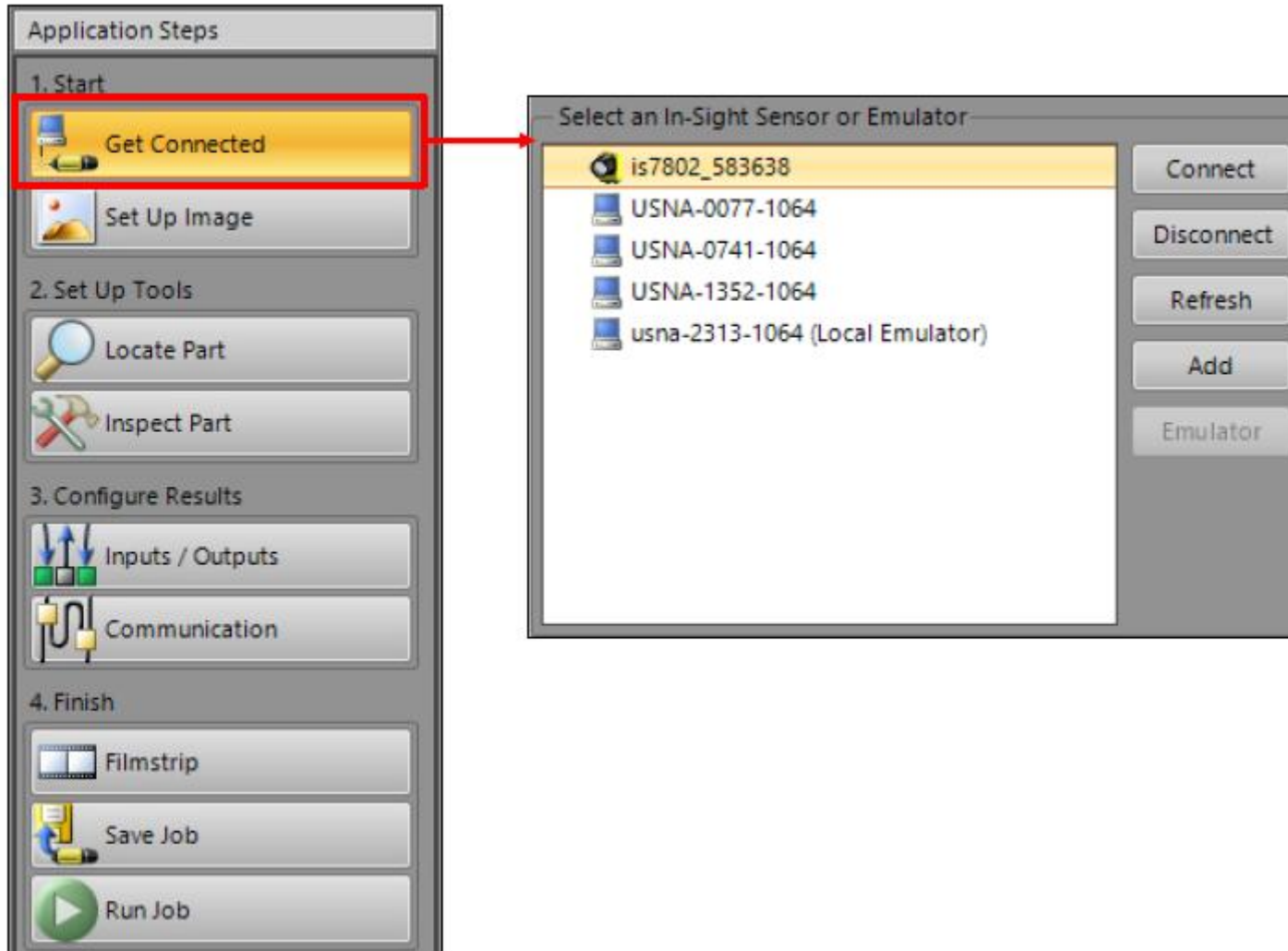
Class	Subnet Mask	Network Address	Host Address
A	255.0.0.0	8 bit	24 bit
B	255.255.0.0	16 bit	16 bit
C	255.255.255.0	24 bit	8 bit

Get Connected



- When you open In-Sight Explorer the default view is **Get Connected**.
- Although there may be more than one sensor or emulator displayed in the list of available tools connected to the network, only one sensor or emulator can be connected.

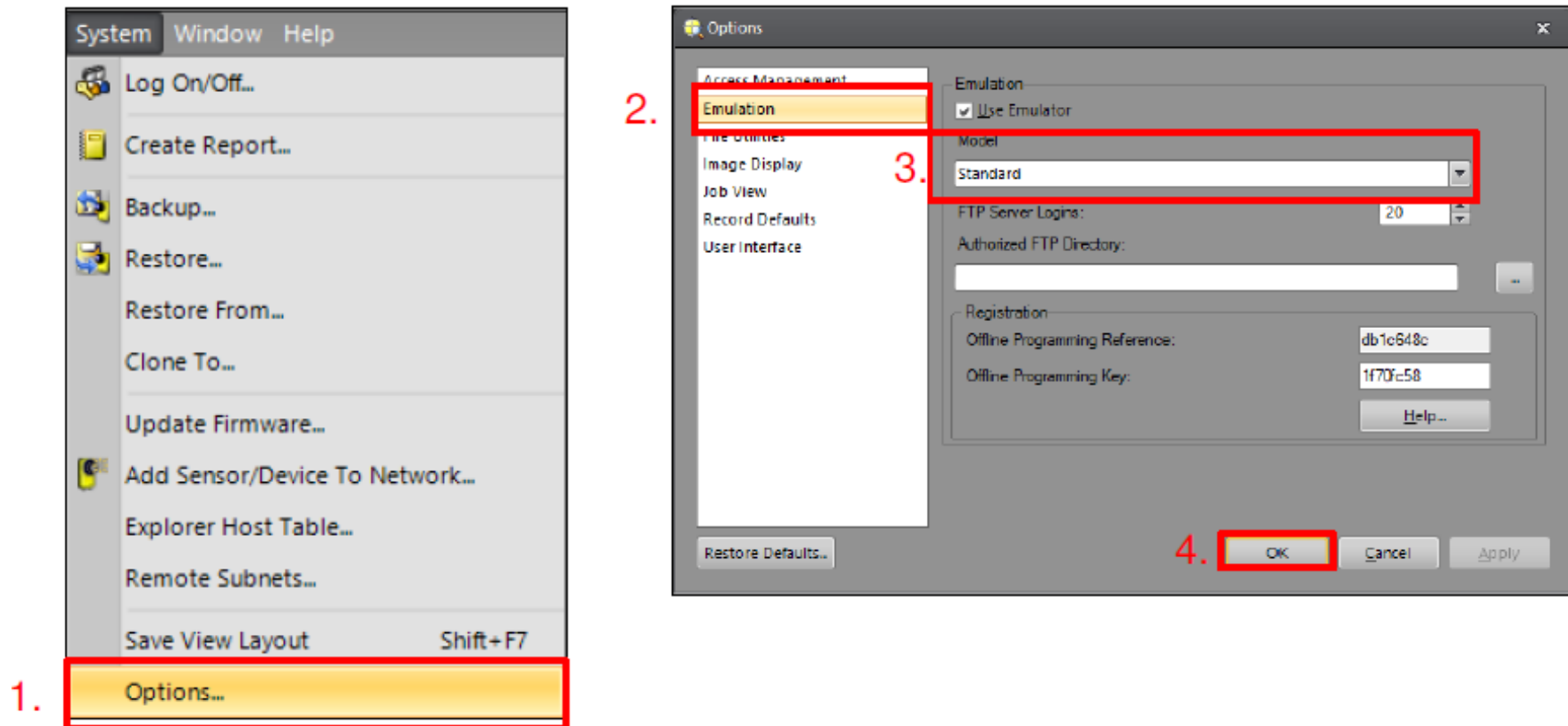
Get Connected



- In-Sight Explorer automatically detects any In-Sight vision systems on your subnet and displays them.
- Vision systems with a **camera icon** represent actual In-Sight vision systems (for example, the In-Sight 7802) while the **computer icon** represent In-Sight emulators running on networked PCs. The name of the emulator is the computer's name under Microsoft Windows.
- To make sure the list is current, select **View - Refresh** or press **<F5>**.

Emulator

- In the System – Options - Emulation menu, In-Sight Explorer can be configured to emulate any model of In-Sight camera.

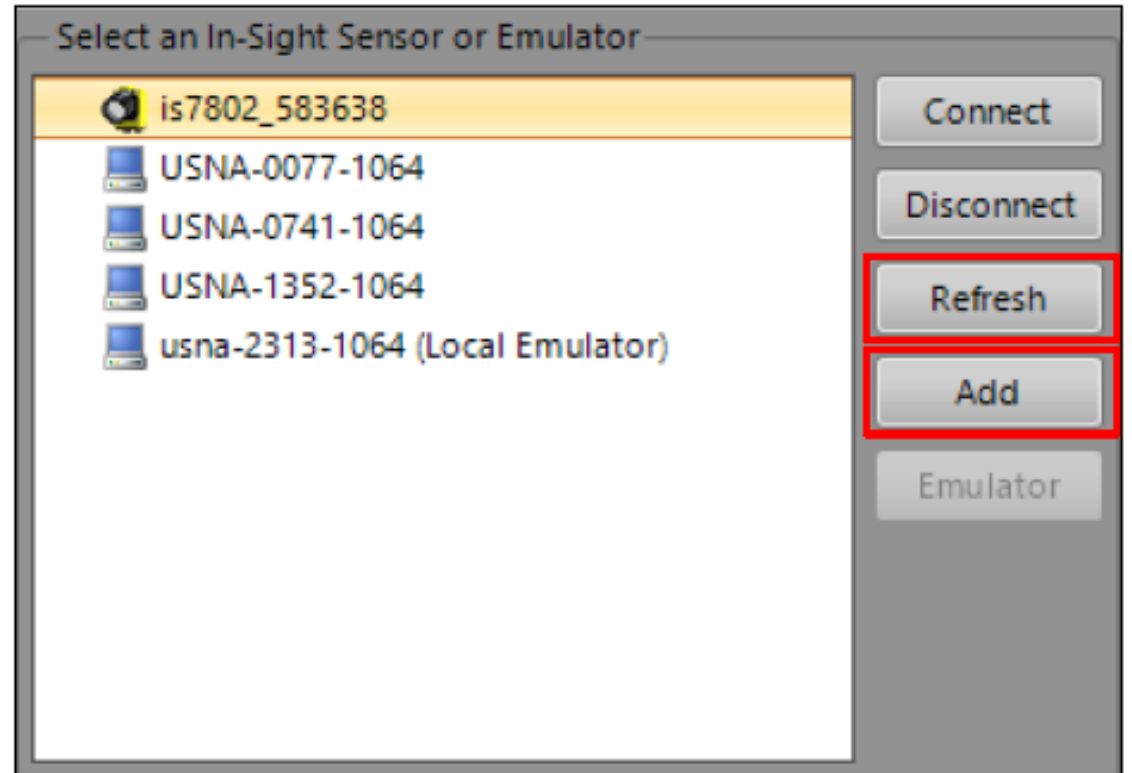


Emulator

- Using an emulator lets you try out cameras that you don't have which may have different tools or different resolution.
- You must use images, and those images must match the resolution of the camera being emulated.
- You can even run the emulator when no cameras are networked to the PC. This is called **standalone mode** or **offline programming**.
- You enable this on a PC by entering a key, which is a number obtained from the Cognex website and is different for each PC. You only need to enter the key once for a given PC.

Add Sensor

- Make sure the camera is powered on and networked (check the LEDs).
- Click the **Refresh** button to make sure the list is current.
- If your camera is still not on the list, click the **Add** button.



Add Sensor

- This automatically detects In-Sight sensors with invalid network settings on your TCP/IP subnet and displays them in a list.
- Any In-Sight sensor or Cognex Ethernet device that is power cycled on the subnet while the dialog is open is also added to the list.

The screenshot shows a software window titled "Add Sensor/Device to Network". At the top, there is a text instruction: "Select an In-Sight sensor or device to add to your network. If the desired sensor/device is not listed, you can add it by cycling its power. Devices may take up to 60 seconds to appear in the list after power is applied." Below this is a "Filter:" label followed by a text input field containing the placeholder "Enter text to filter the list." The main area of the window is a table with four columns: "Host Name", "Type", "MAC", and "IP". The table is currently empty. To the right of the table, there are network configuration options. A "Host Name:" label is followed by a text input field. Below that are two radio buttons: "Obtain IP Address Automatically (DHCP)" (which is selected) and "Use The Following Network Settings". Under the second option, there are five text input fields for "IP Address:", "Subnet Mask:", "Default Gateway:", "DNS Server:", and "Domain Name:". Below these fields is a button labeled "Copy PC Network Settings". At the bottom right, there are two unchecked checkboxes: "Reset Admin Password" and "Reset Sensor Settings to Factory Defaults". At the bottom of the window, there are four buttons: "Flash Lights", "Refresh", "Show All" (with a radio button), and "Show New" (with a radio button). On the far right, there are "Apply" and "Close" buttons.

Add Sensor

- Cameras with incorrect network settings should appear in the **Add Sensor** dialog box.
- Click on the **sensor**, change the settings to match the network, and click the **Apply** button. If the camera still appears in the dialog box, keep the box open and recycle the power on the camera.

Add Sensor/Device to Network

Select an In-Sight sensor or device to add to your network. If the desired sensor/device is not listed, you can add it by cycling its power. Devices may take up to 60 seconds to appear in the list after power is applied.

Filter:

Host Name	Type	MAC	IP
is7802_5...	7802	00-d0-24-58-36-38	169.254.0.5

Host Name:

☒ Obtain IP Address Automatically (DHCP)
☐ Use The Following Network Settings

IP Address:
Subnet Mask:
Default Gateway:
DNS Server:
Domain Name:

☐ Reset Admin Password
☐ Reset Sensor Settings to Factory Defaults

☐ Show All ☐ Show New

Add Sensor

- To see and change models already on the network:
 - Click the **Show All** radio button
 - Now you can select one and alter its settings

Host Name	Type	MAC	IP
cioMicro_13366f	CIO-MICRO	00-d0-24-13-36-6f	10.11.80.211
cioMicro_4fd41e	CIO-MICRO	00-d0-24-4f-d4-1e	10.11.80.35
IS2000_Table	2000-130	00-d0-24-42-76-...	10.11.80.201
is2000-130_PJC	2000-130	00-d0-24-42-d5-5c	10.11.80.206
is2000-130C_AbbysBabby	2000-130C	00-d0-24-45-65-5a	10.11.80.95
is2000-130C_Training	2000-130C	00-d0-24-45-6a-10	10.11.80.227
is2000-Romeos_Desk	2000-130C	00-d0-24-44-e3-6e	10.11.80.230
is5100_12fa76	5100	00-d0-24-12-fa-76	10.11.0.238
is5600_must_proceed_hastilyo	5600	00-d0-24-1e-f7-48	10.11.0.40
is7402_numbahfour	7402	00-d0-24-1b-fe-b8	10.11.1.15
is7402C-DashboardCam	7402C	00-d0-24-1b-80-...	10.11.81.10
is7402-Romes_Desk	7402	00-d0-24-1b-f3-0a	169.254.0.1
is7802_3d7188	7802	00-d0-24-3d-71-...	10.11.80.11
is7802_flerg	7802	00-d0-24-3d-73-...	10.11.80.57
TestAdvantage102	CIO-MICRO	00-d0-24-40-d2-c6	10.11.0.121
TS_1500_1_theINJUSTICE	ism1500	00-d0-24-21-12-...	10.11.80.54
TS_5705	5705	00-d0-24-22-8b-...	10.11.0.55
TS_5705C_babycam	5705C	00-d0-24-23-1c-94	10.11.80.220
TS_7010_NOT_STEVE	7010	00-d0-24-1b-88-...	10.11.80.19
TS_7200_1_PatTest8-12	7200	00-d0-24-13-9c-ea	169.254.8.84
TS_7200_SCWABBZ	7200	00-d0-24-1b-f2-34	10.11.80.34
TS_7200C_Fiyve	7200C	00-d0-24-18-f8-06	10.11.82.95
TS_7402_PJC	7402	00-d0-24-1b-7a-...	10.11.80.246
TS_CIO_Jer_StaticShock	CIO-MICRO	00-d0-24-13-35-...	10.11.0.177
TS_CIO_MICRO_CC	CIO-MICRO-...	00-d0-24-11-4c-e4	10.11.80.242
TS_PB_CIO_MICRO_133905	CIO-MICRO	00-d0-24-13-39-...	10.11.80.40

Host Name: is2000-130C_Training

☒ Obtain IP Address Automatically (DHCP)

☐ Use The Following Network Settings

IP Address: 10 , 11 , 80 , 227

Subnet Mask: 255 , 255 , 0 , 0

Default Gateway: 10 , 11 , 205 , 205

DNS Server: 10 , 10 , 160 , 1

Domain Name: pc.cognex.com

Copy PC Network Settings

☐ Reset Admin Password

☐ Reset Sensor Settings to Factory Defaults

Flash Lights Refresh ☒ Show All ☐ Show New Apply Close

Add Sensor

- Resetting a Camera to Factory Defaults:
 - Resets the discrete I/O, Startup, and User List settings.
 - Releases the IO module.
 - Does not affect jobs in the camera – but it is a good idea to have a Backup.

Add Sensor/Device to Network

Select an In-Sight sensor or device to add to your network. If the desired sensor/device is not listed, you can add it by cycling its power. Devices may take up to 60 seconds to appear in the list after power is applied.

Filter:

Host Name	Type	MAC	IP
is7802_5...	7802	00-d0-24-58-36-38	169.254.0.5

Host Name:

☐ Obtain IP Address Automatically (DHCP)

☐ Use The Following Network Settings

IP Address:

Subnet Mask:

Default Gateway:

DNS Server:

Domain Name:

☐ Reset Admin Password

☒ Reset Sensor Settings to Factory Defaults

☐ Show All ☐ Show New

Online vs. Offline

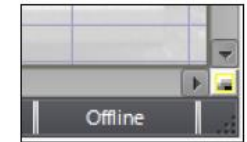
- **Online** means that all In-Sight Input and Output signals (discrete, serial, network, and non-manual triggers) are enabled.
- When Online, you can do this:
 - Acquisition triggers
 - Serial I/O
 - Discrete I/O
 - Network I/O
- But not this:
 - Edit spreadsheet
 - Open Property Sheets



Online means that all In-Sight Input and Output signals are enabled.



Offline means that most In-Sight Input and Output signals are disabled.



Online vs. Offline

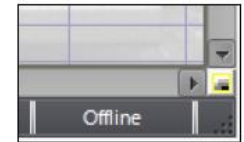
- **Offline** means that most In-Sight Input and Output signals are disabled.
- When Offline, you can do this:
 - Edit spreadsheet
 - Open Property Sheets
- But not this:
 - Acquisition triggers
 - Serial I/O
 - Discrete I/O
 - Network I/O



Online means that all In-Sight Input and Output signals are enabled.

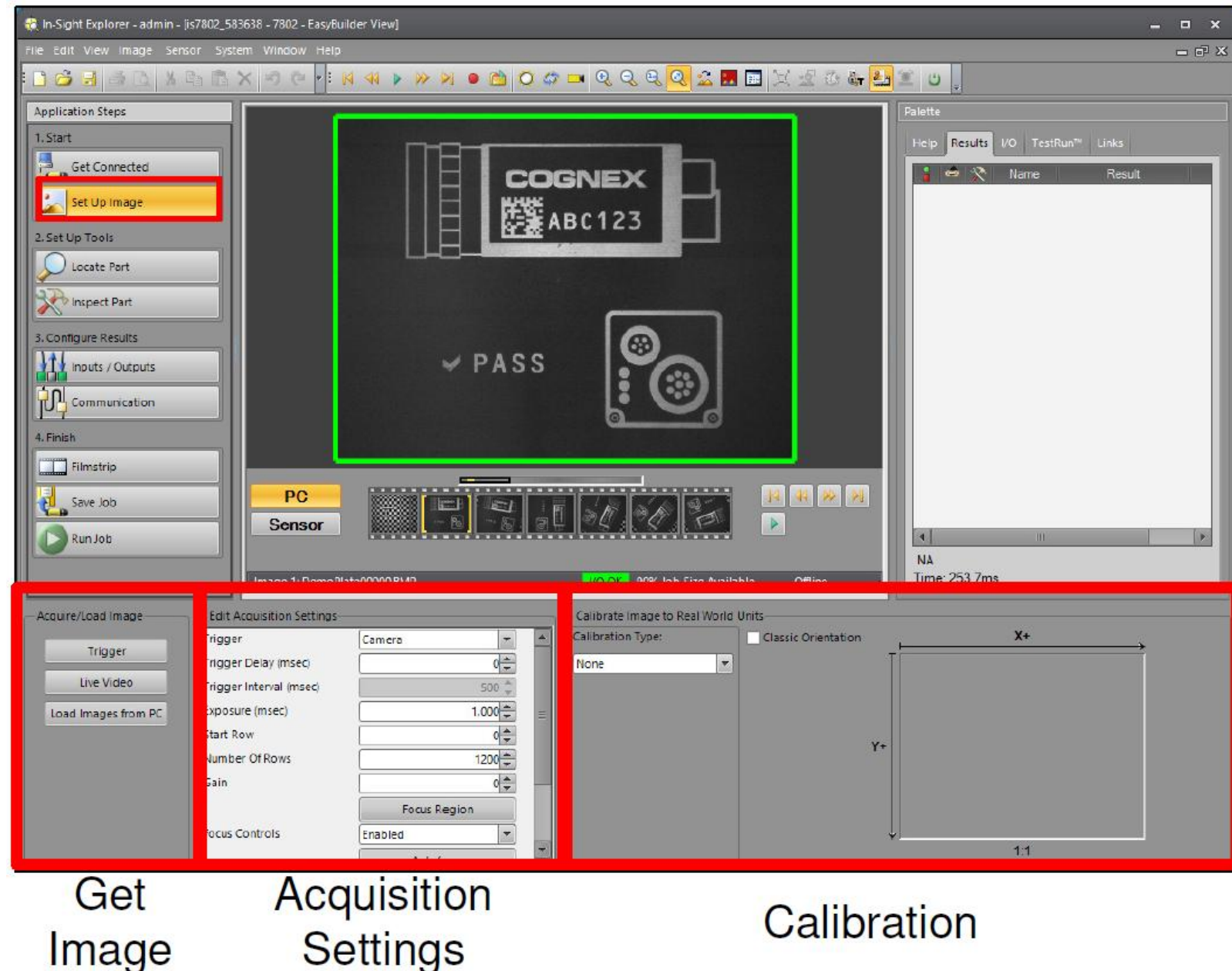


Offline means that most In-Sight Input and Output signals are disabled.



Set Up Image

- We are already connected to the camera; the next step is to **Set up the Image**.
 - **Get Image** – Where is the image being captured?
 - **Acquisition Settings** – How is the image being acquired? This is where the trigger, exposure, light and focus can be controlled.
 - **Calibration** – A critical operation in precision alignment applications. Errors in calibration can have subtle but important effects on results.



Acquire / Load Image

- The **Acquire / Load Image** group box allows you to instantly:
 - Acquire an image – single shot (manual acquisition)
 - Display live video
 - Load previously acquired images from a PC

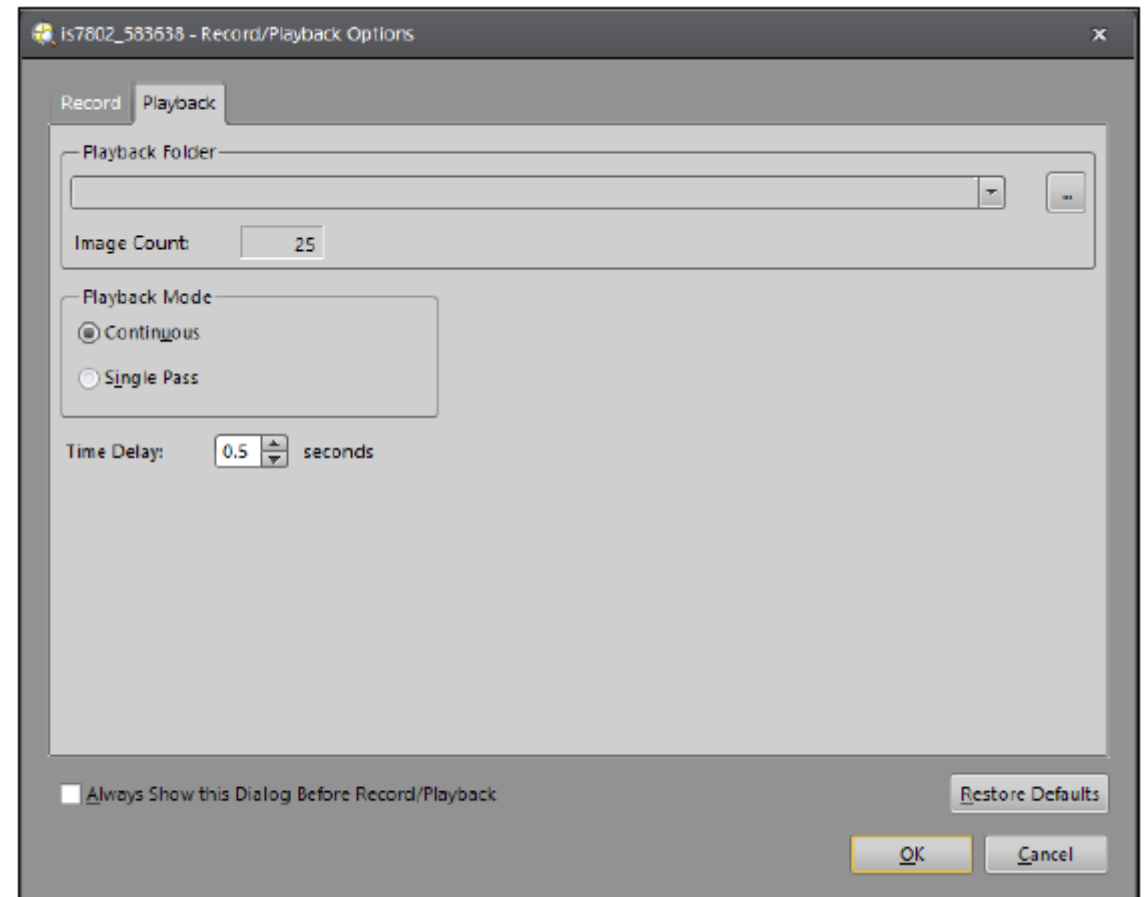
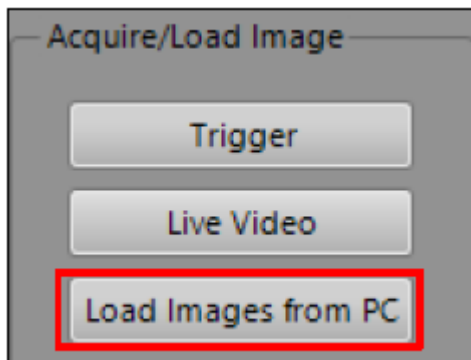
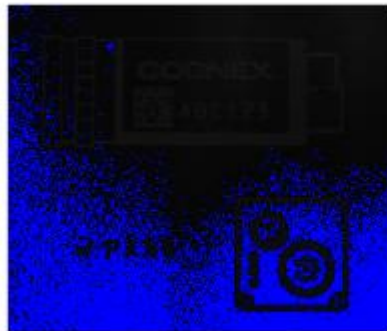


Image Saturation Tool

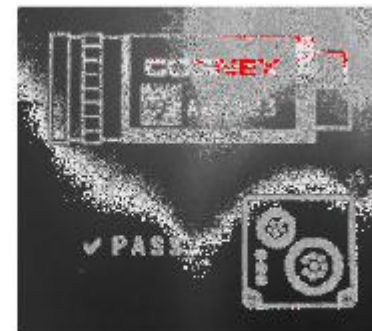
- **Show Brightness Feedback** toolbar button assists with obtaining a good image.
- The goal is not to have blue or red span a feature and background.



Not bright enough



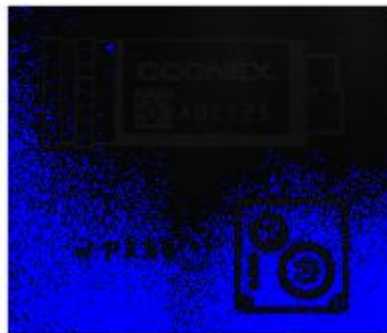
Too bright



Good

Image Saturation Tool

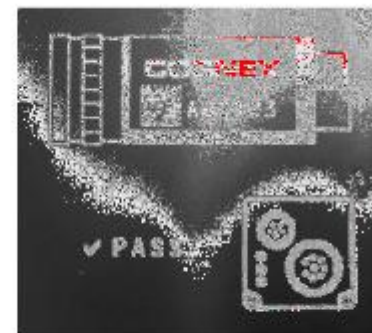
- Red or blue shown on the image is not necessarily a problem if it is limited to the background or limited to a feature. It can indicate a problem when it overlaps a feature and the background.
- The leftmost image needs to have more brightness. This can be accomplished with more lighting, a larger lens aperture, and/or a longer exposure.



Not bright enough



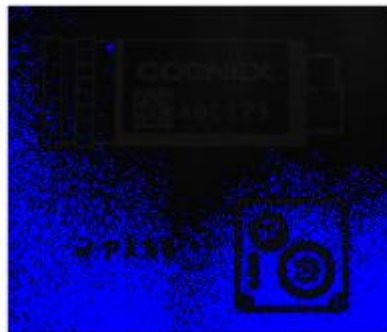
Too bright



Good

Image Saturation Tool

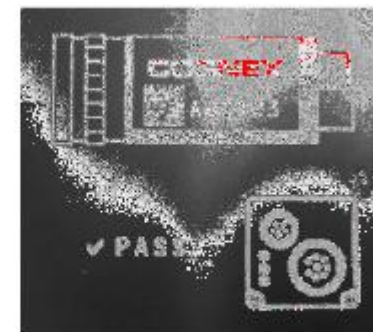
- The middle image needs to have less brightness. This can be accomplished with less lighting, a smaller lens aperture, and/or a shorter exposure.
- The rightmost image is good. Even though there is some red, it is limited to features and does not overlap into the background.
- The grayscale threshold is set in *System* → *Options*, with default values of 5 (pixels below 5 are displayed in blue) and 240 (pixels above 240 are displayed in red).



Not bright enough



Too bright



Good

Acquisition Settings - Trigger

Edit Acquisition Settings

Trigger: Camera

Trigger Delay (msec): 0

Trigger Interval (msec): 500

Exposure (msec): 9.000

Start Row: 0

Number Of Rows: 1200

Gain: 0

Focus Region

Focus Controls: Enabled

Autofocus

Focus Position: 136

Save Focus Position with Job: ☒

Light Settings

Camera

Continuous

External

Manual

Network

Industrial Ethernet

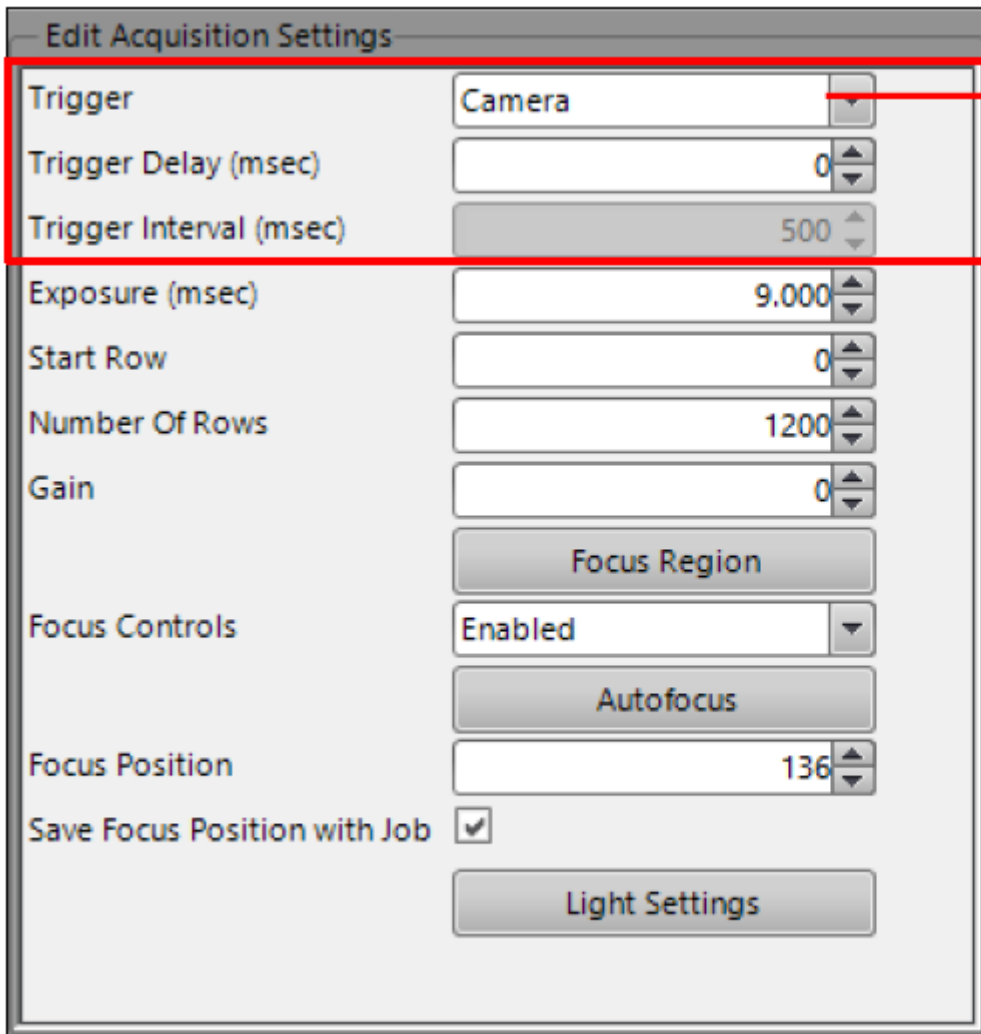
- The **Edit Acquisition Settings** group box allows you to:

- Adjust the acquisition trigger controls
- Compensate for exposure and brightness
- Control lights for In-Sight cameras

- **Trigger** for Online mode

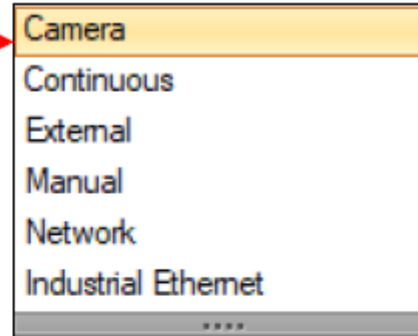
- **Camera** – Trigger (+) and Trigger (-) connections. This is hardwiring trigger.
- **Continuous** – Enables free running image acquisitions (pseudo-live)
- **External** – Trigger via discrete input (via IO board)
- **Manual** – Enables image acquisition when pressing <F5>

Acquisition Settings - Trigger



Edit Acquisition Settings

Trigger	Camera
Trigger Delay (msec)	0
Trigger Interval (msec)	500
Exposure (msec)	9.000
Start Row	0
Number Of Rows	1200
Gain	0
	Focus Region
Focus Controls	Enabled
	Autofocus
Focus Position	136
Save Focus Position with Job	☒
	Light Settings



Camera
Continuous
External
Manual
Network
Industrial Ethernet

- **Trigger Interval** - Allows you to specify an interval in milliseconds (0 to 10,000; default is 500), between acquisitions when the trigger control is set to Continuous.
- **Trigger Delay** - Allows you to specify a delay, in milliseconds (0 to 10,000), between the time the trigger is received and the time the In-Sight vision system begins the acquisition.
- The Trigger control must be in either Camera or Network to enable this functionality.

Acquisition Settings - Exposure

- **Exposure** - Allows you to specify the exposure time. When the In-Sight vision system receives a trigger signal, light is integrated in the image sensor array for the specified duration.

The screenshot shows the 'Edit Acquisition Settings' dialog box with the following settings:

Setting	Value
Trigger	Camera
Trigger Delay (msec)	0
Trigger Interval (msec)	500
Exposure (msec)	9.000
Start Row	0
Number Of Rows	1200
Gain	0
Focus Region	Focus Region
Focus Controls	Enabled
Autofocus	Autofocus
Focus Position	136
Save Focus Position with Job	☒
Light Settings	Light Settings

Image Sensor Transfer Controls

- The **Image Sensor Transfer Controls** allow you to define at what row of the image sensor the image will begin to be transferred into memory, and how many rows of the image sensor will be used to construct the image.
- **Start Row** defines the first row to be transferred from the image sensor into memory on the In-Sight vision system.
- **Number of Rows** defines the number of rows to be transferred into memory on the In-Sight vision system.

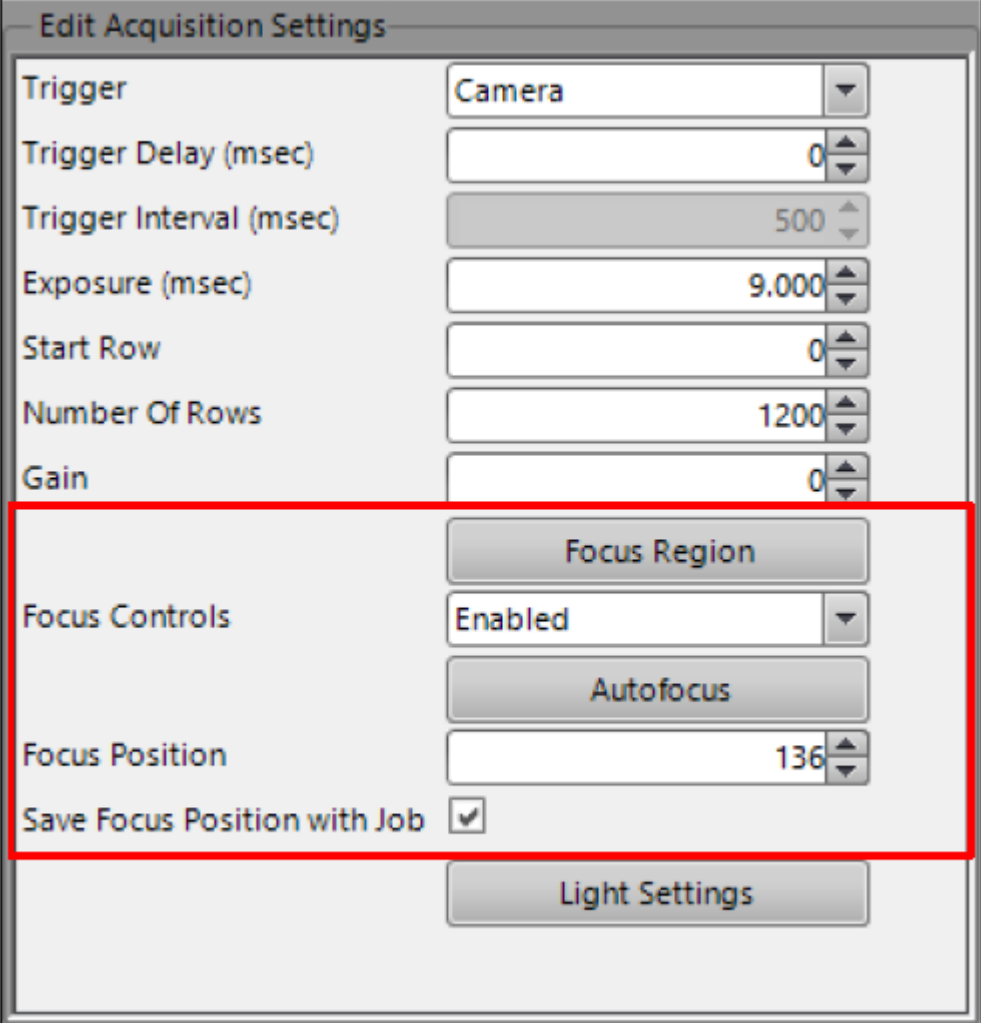
The screenshot shows the 'Edit Acquisition Settings' dialog box with the following settings:

Parameter	Value
Trigger	Camera
Trigger Delay (msec)	0
Trigger Interval (msec)	500
Exposure (msec)	9.000
Start Row	0
Number Of Rows	1200
Gain	0
Focus Controls	Enabled
Focus Position	136
Save Focus Position with Job	☒

Buttons visible: Focus Region, Autofocus, Light Settings.

Focus Controls

- The **Focus Controls** are used to control the focus position of the lens on supported In-Sight 7000 series vision systems, by either automatically focusing the lens or moving the lens to a specific focus position.
- **NOTE:** *The Focus Controls only work when the vision system is offline.*
- **Focus Controls** defines whether the Focus Controls are enabled. By default, if a sensor has a valid default focus position, the Focus Controls are enabled, if a focus position has not been established, the Focus Controls are disabled.



The screenshot shows the 'Edit Acquisition Settings' dialog box. The 'Focus Controls' section is highlighted with a red rectangle. This section includes a 'Focus Region' button, a 'Focus Controls' dropdown menu set to 'Enabled', an 'Autofocus' button, a 'Focus Position' spinner set to 136, and a 'Save Focus Position with Job' checkbox which is checked. Other settings visible in the dialog include 'Trigger' (Camera), 'Trigger Delay (msec)' (0), 'Trigger Interval (msec)' (500), 'Exposure (msec)' (9.000), 'Start Row' (0), 'Number Of Rows' (1200), and 'Gain' (0). A 'Light Settings' button is located at the bottom right of the dialog.

Setting	Value
Trigger	Camera
Trigger Delay (msec)	0
Trigger Interval (msec)	500
Exposure (msec)	9.000
Start Row	0
Number Of Rows	1200
Gain	0
Focus Region	Focus Region
Focus Controls	Enabled
Autofocus	Autofocus
Focus Position	136
Save Focus Position with Job	☒
Light Settings	Light Settings

Focus Controls

- **Autofocus Button** automatically focuses the lens, to maximize image sharpness within the region.
- **Focus Position** moves the lens to a specific focus position and acquires a new image.
- **Save Focus Position with Job** defines whether the focus position will be saved with the job and applied to the lens when the job is reloaded. By default, the checkbox is checked.

The screenshot shows the 'Edit Acquisition Settings' dialog box. The 'Focus Controls' section is highlighted with a red rectangle. It includes the following settings:

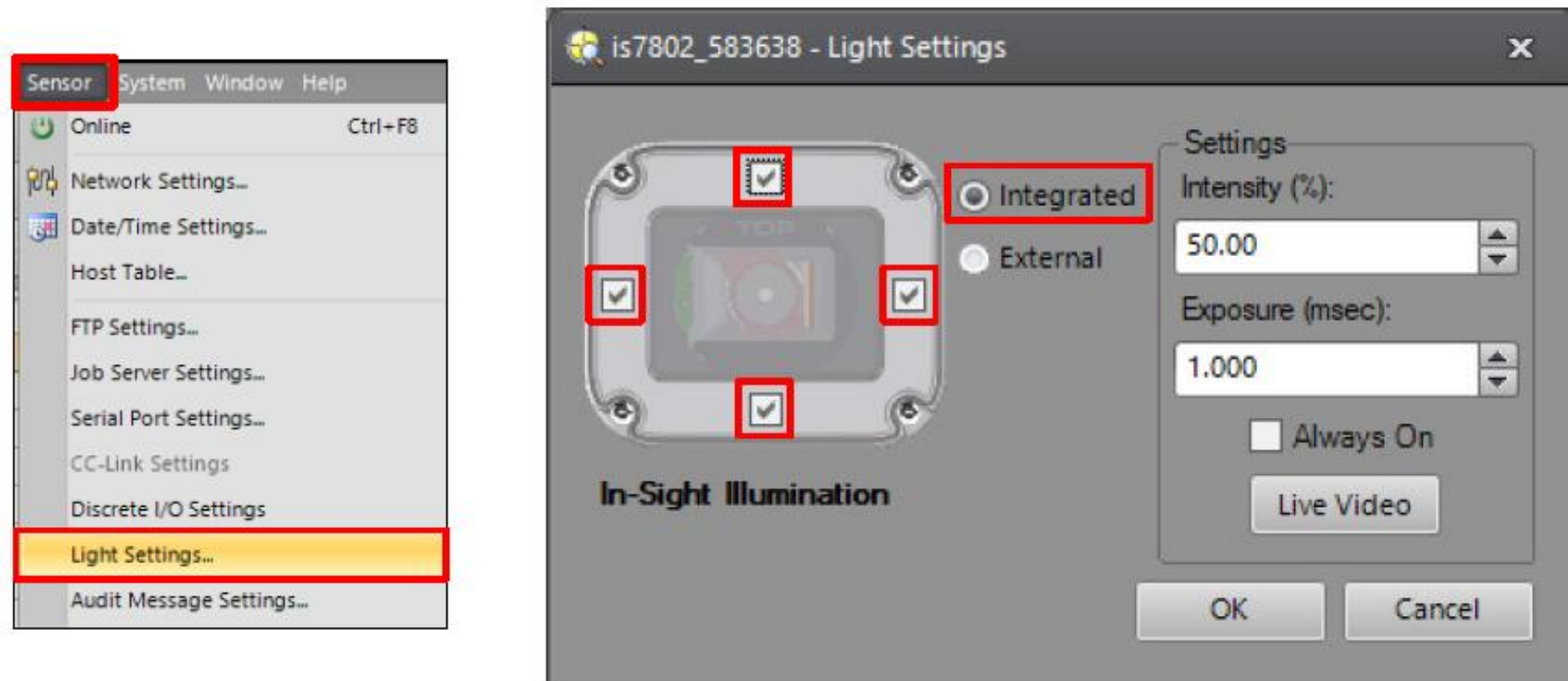
- Focus Region**: A button to select the focus region.
- Focus Controls**: A dropdown menu set to 'Enabled'.
- Autofocus**: A button to initiate autofocus.
- Focus Position**: A numeric input field set to 136.
- Save Focus Position with Job**: A checked checkbox.

Other settings visible in the dialog include:

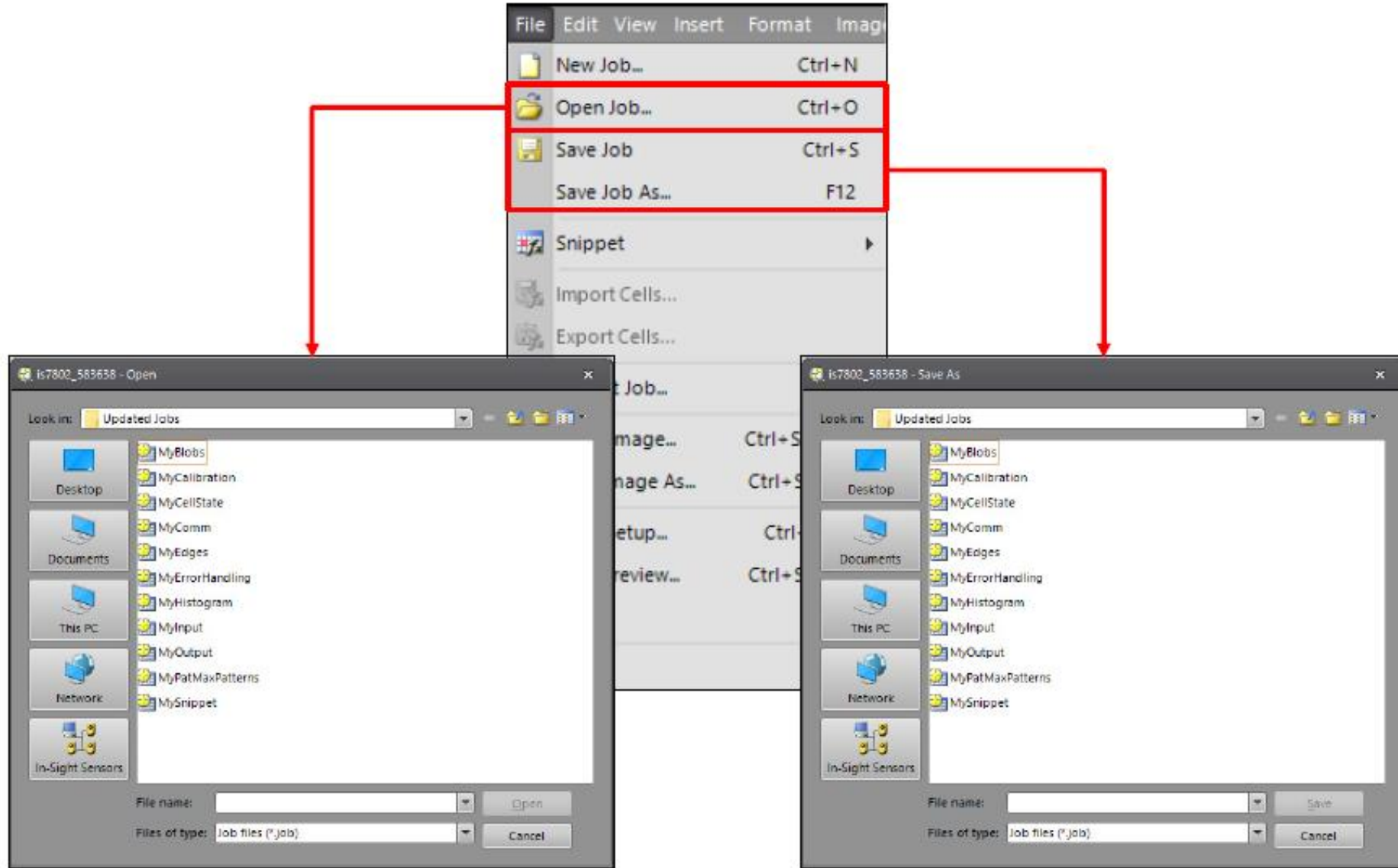
- Trigger**: Camera
- Trigger Delay (msec)**: 0
- Trigger Interval (msec)**: 500
- Exposure (msec)**: 9.000
- Start Row**: 0
- Number Of Rows**: 1200
- Gain**: 0
- Light Settings**: A button at the bottom.

In-Sight Integrated Light

- This dialog allows you to configure 0-4 banks of lights. Check on any checkboxes at the left to activate that banks. Determine the proper **Exposure** and **Intensity** values for good images.



Open and Save Jobs

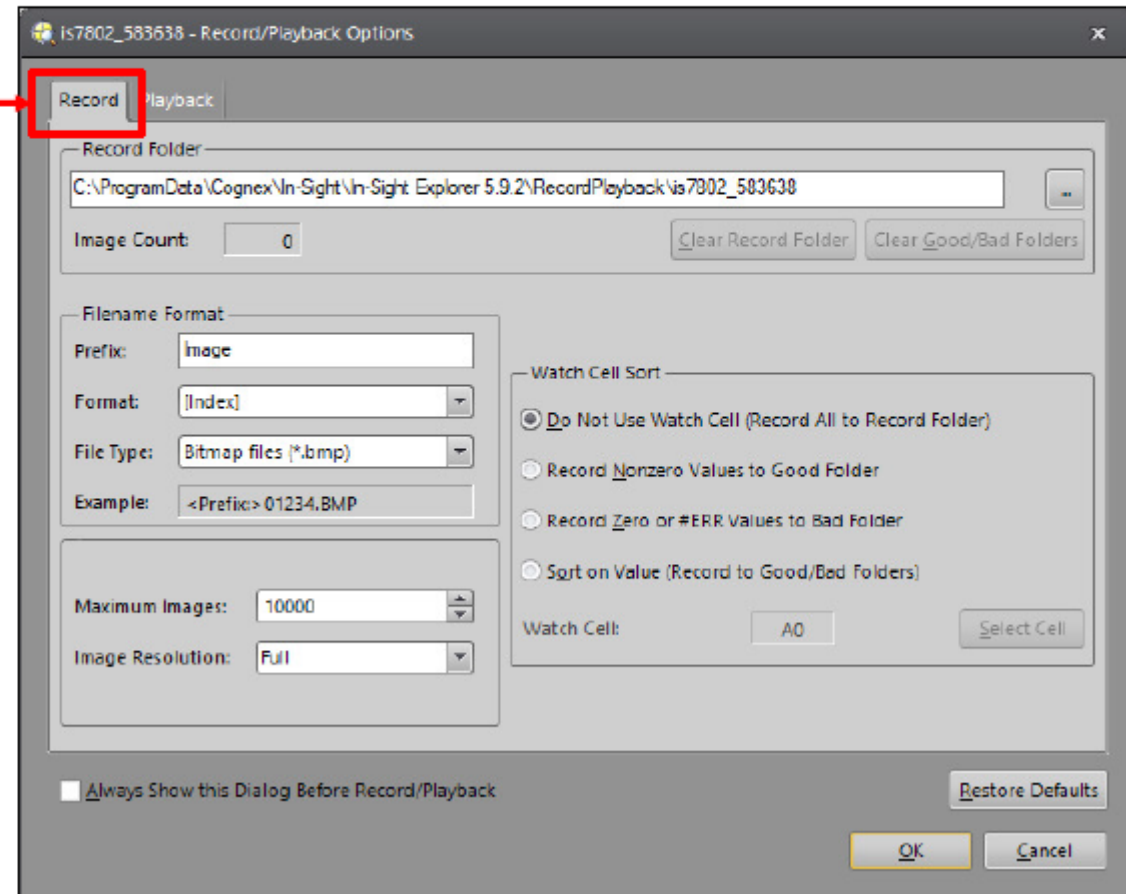


Recording Images

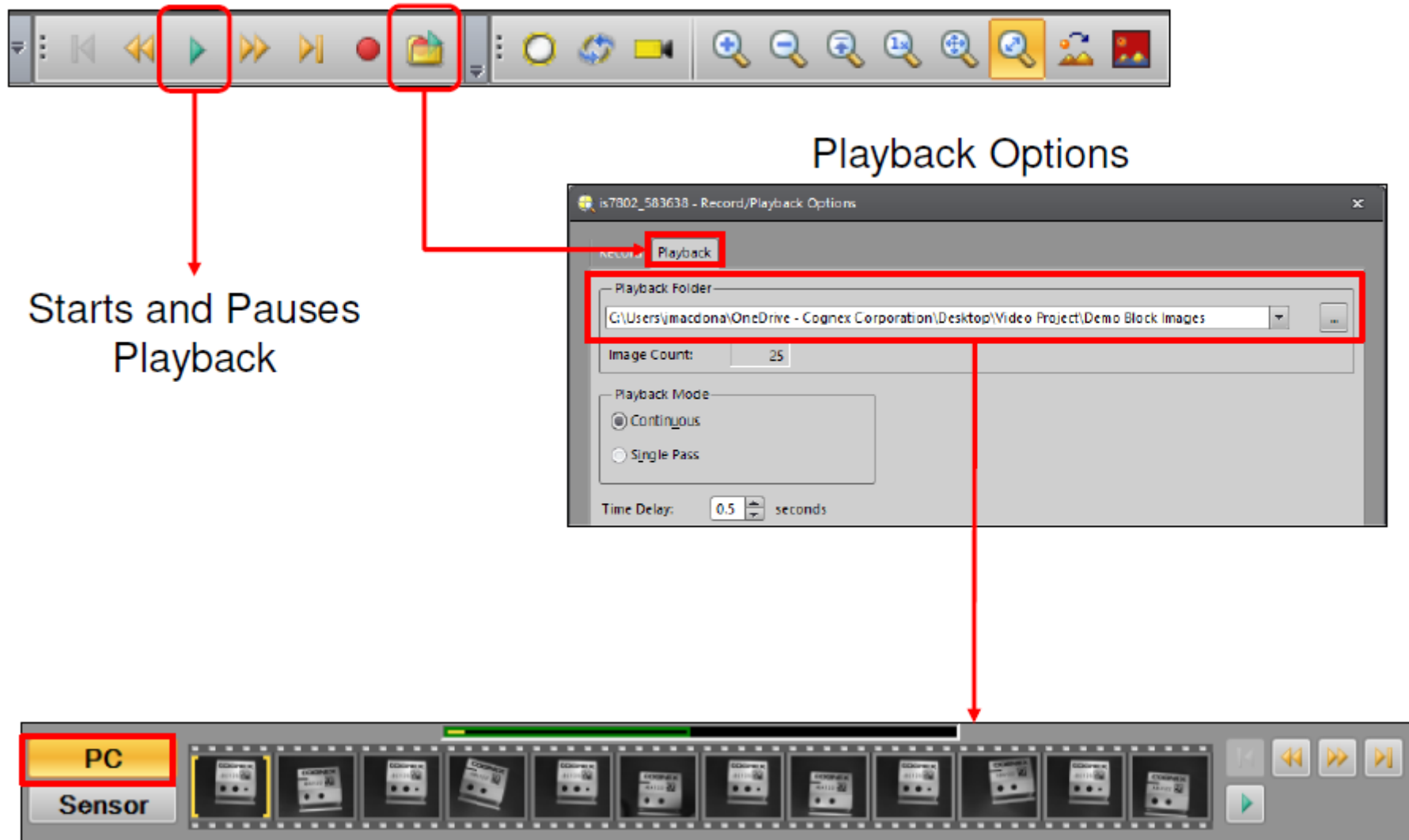


Record Options

Starts and Stops
Recording



Recording Images



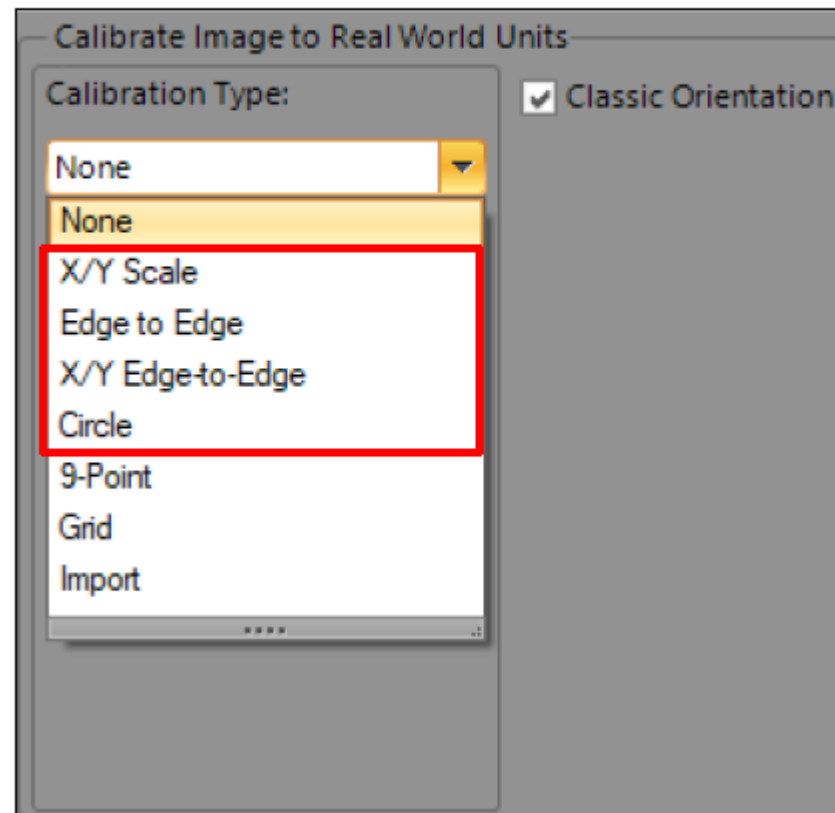
Calibration - From Pixels to Real-World Measurements

- **Pixels** are photodetectors on an image sensor that convert light into electrical signals.
- When a camera captures an image of a part, the dimensions of the part **measured in pixels** are called **image dimensions**.
- For example, the length of the part may be **108 pixels**.
- However, what we are really interested in is the **physical dimension** of the part. For example, the actual length of the part may be **100 mm**.
- So, how do we find the relationship between the **pixel coordinate system** and the **real-world coordinate system**?
- This is achieved through **calibration**.
- **What affects the pixel-to-real-world relationship?**
 - **Distance** between the camera and the part
 - **Camera mounting** — directly above or at an angle
 - **Optics / lens** — field of view and magnification
 - **Perspective and lens distortion**

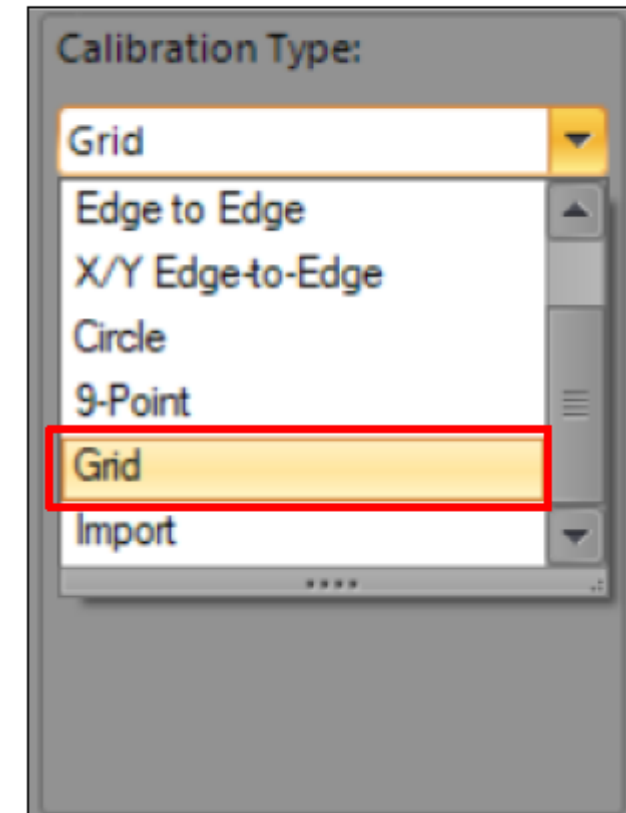
Calibration

- There are two types of calibration in EasyBuilder:
 1. In **Scale Calibration**, you identify a distance on the image and tell EasyBuilder what that distance is in real-world units, such as mm. EasyBuilder already knows what the distance is in terms of pixels, so it can then convert the distance from pixels to real-world units.

Scale Calibration



Grid Calibration

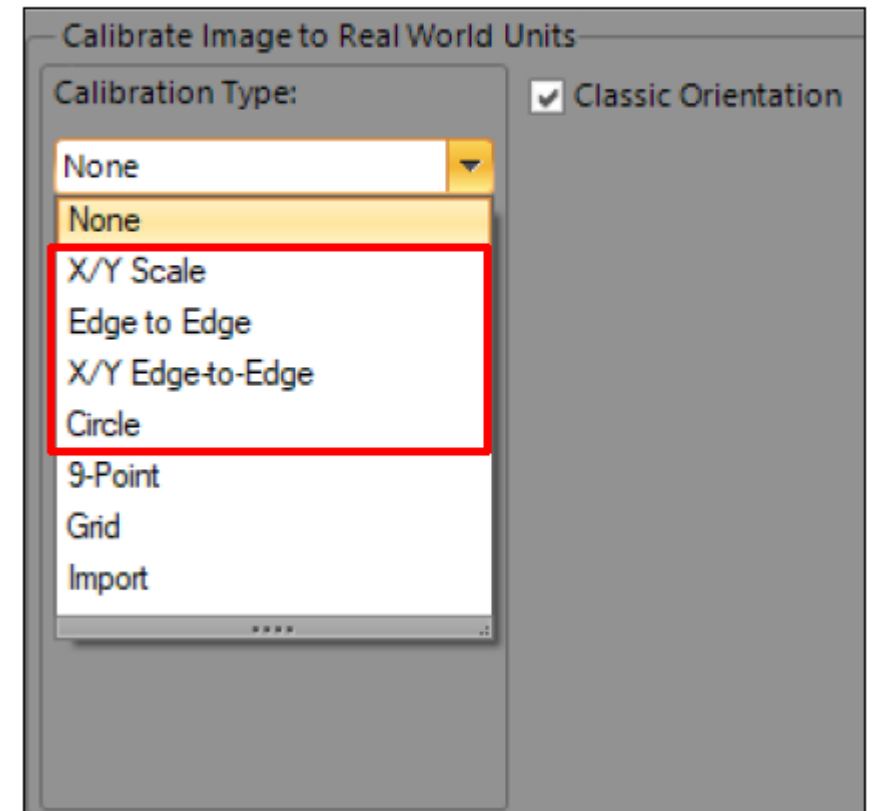


Scale Calibration

Scale Calibration is useful in applications where you are inspecting a part or object and you know the real-world measurement of the part or object. It does not account for radial, barrel, or perspective distortion.

- The **X/Y Scale** is used when you know the X and Y scale values (e.g., mm/pixel) for calculating a simple calibration.
- **Edge to Edge** is used when you know the distance between two edge features.
- **X/Y Edge-to-Edge** is used when you know the distance between **two edge pairs**; one in the horizontal direction, and another in the vertical direction. Combining the horizontal measurement and dimension with the vertical measurement and dimension creates two independent scales.
- **Circle** is used when you know the diameter of a circular feature.

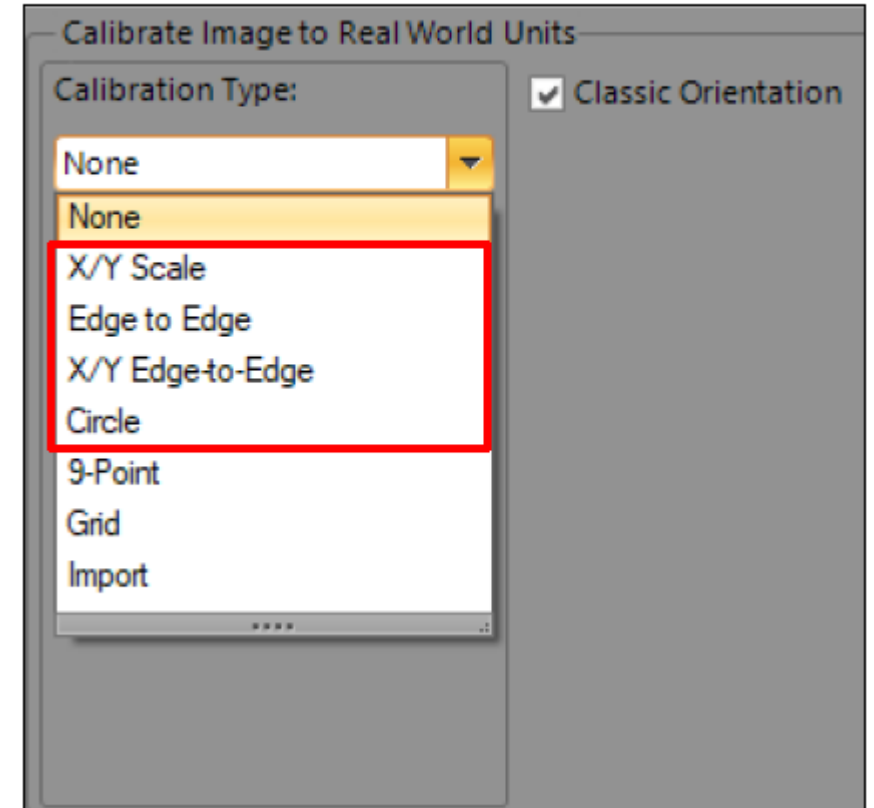
Scale Calibration



Scale Calibration

- The **Scale Calibration** options do not account for lens distortion, therefore, for the greatest degree of accuracy, ensure that the In-Sight system is mounted perpendicular to the inspected surface.
- Otherwise, consider using **Grid Calibration** to remove the distortion and more accurately calibrate your image.

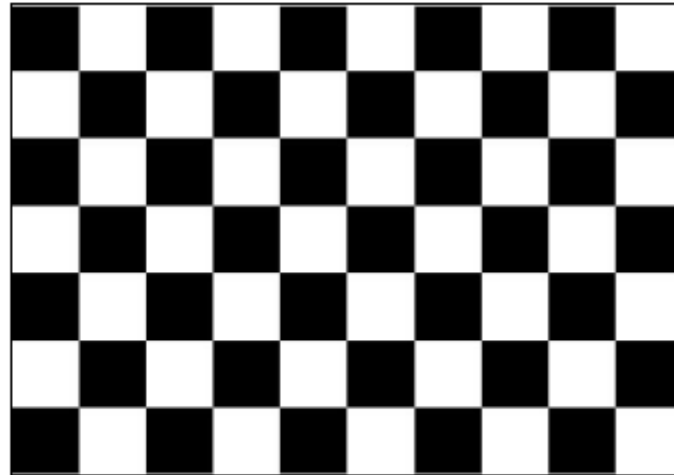
Scale Calibration



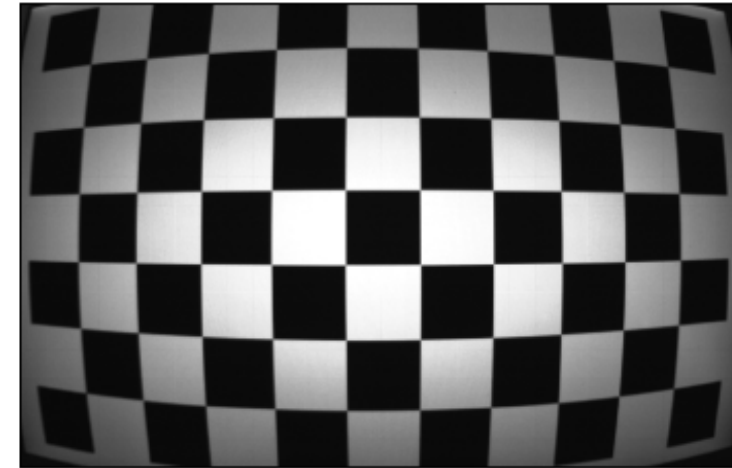
Lens Distortion

- Lenses have distortion effects on the image because the curved glass elements inside them refract, or bend, light in slightly different ways, leading to a failure to perfectly render straight lines, especially with wide-angle lenses.

Pincushion	Wave	Barrel	Actual Part
			



Undistorted
Image



Barrel
Distortion

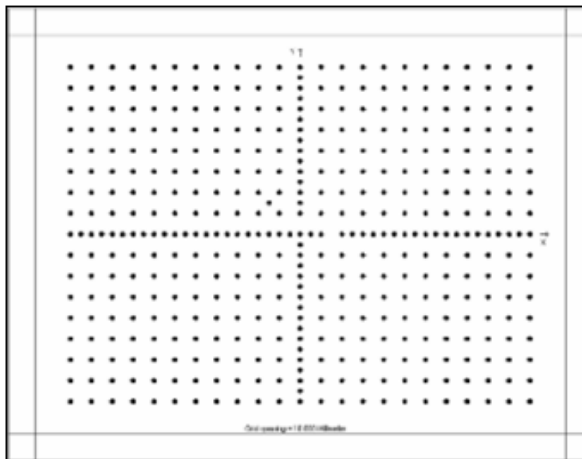
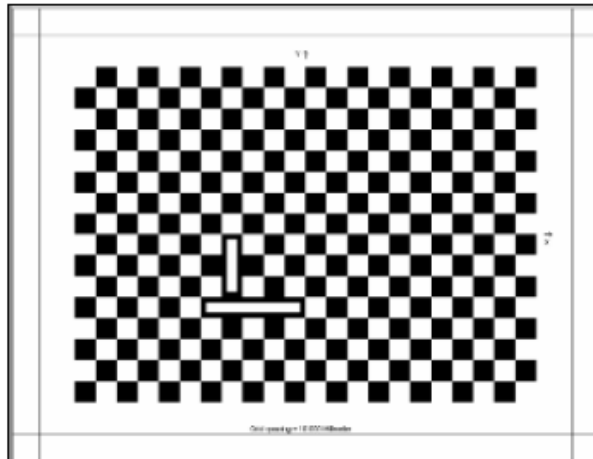
Lens Distortion

- Cameras that are not mounted perpendicular to the surface exhibit perspective distortion. Objects' dimensions are skewed depending on the angle and location.

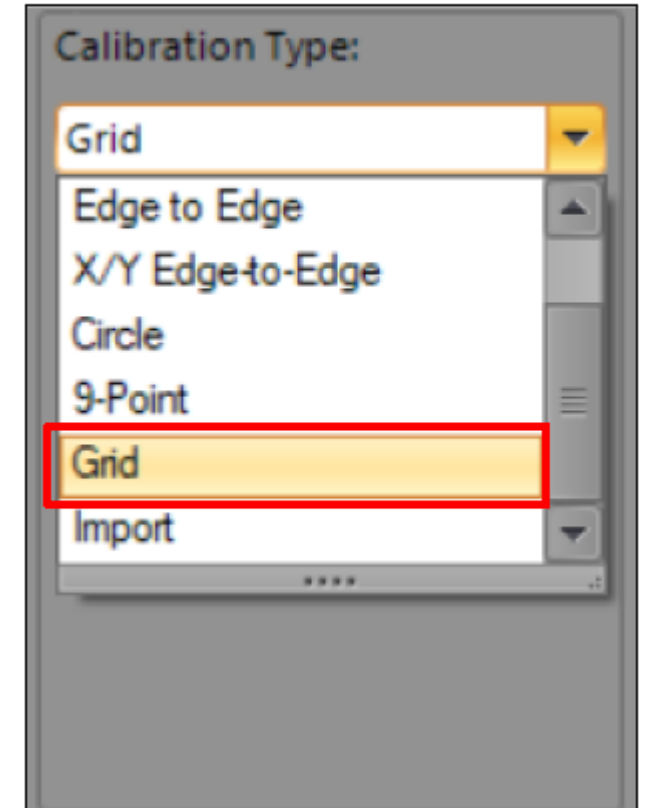


Grid Calibration

2. In **Grid Calibration**, you place a standard grid pattern (dots or squares) under the camera and tell EasyBuilder what the real-world distance is between dots or the size of the checkers. EasyBuilder locates all the dots or corners of squares and creates a table that relates each distance in pixels to its corresponding distance in real-world units. That table is stored in the job.

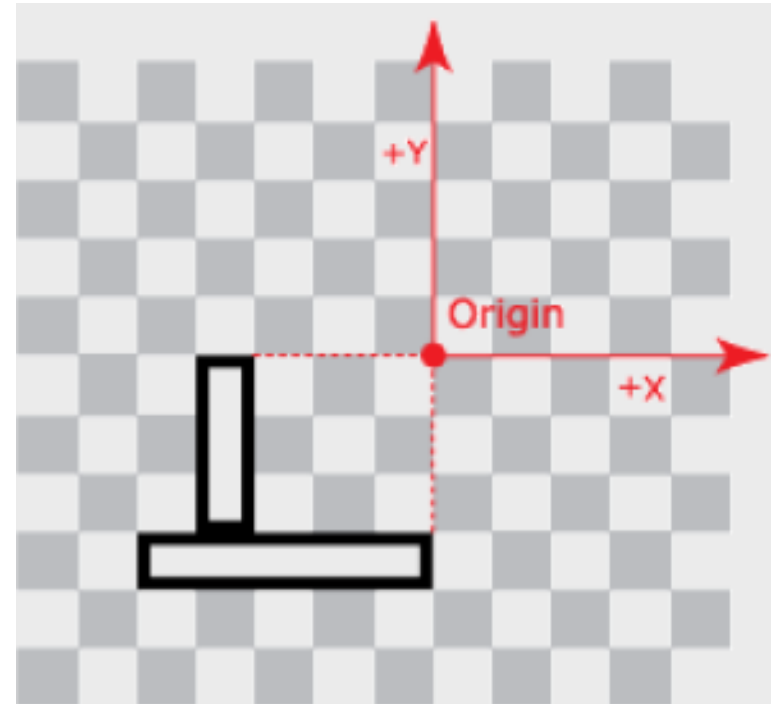


Grid Calibration



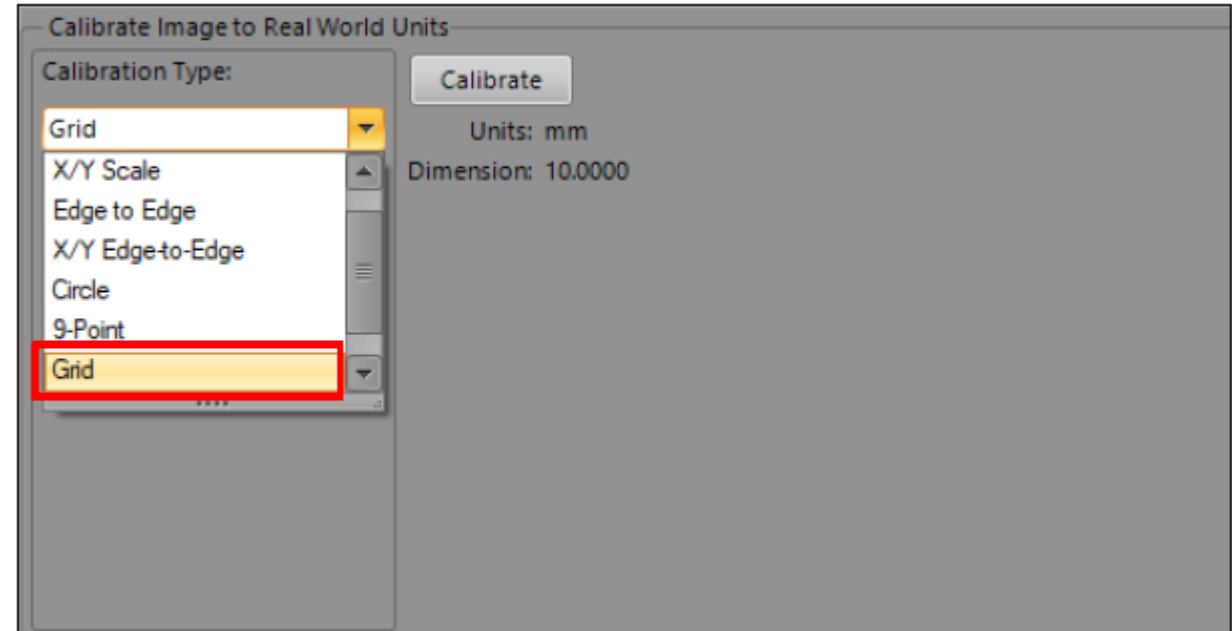
Grid Calibration

- The fiducial for a checkerboard grid is the pattern of two rectangles. It defines the origin (0,0) and the angle of the grid.



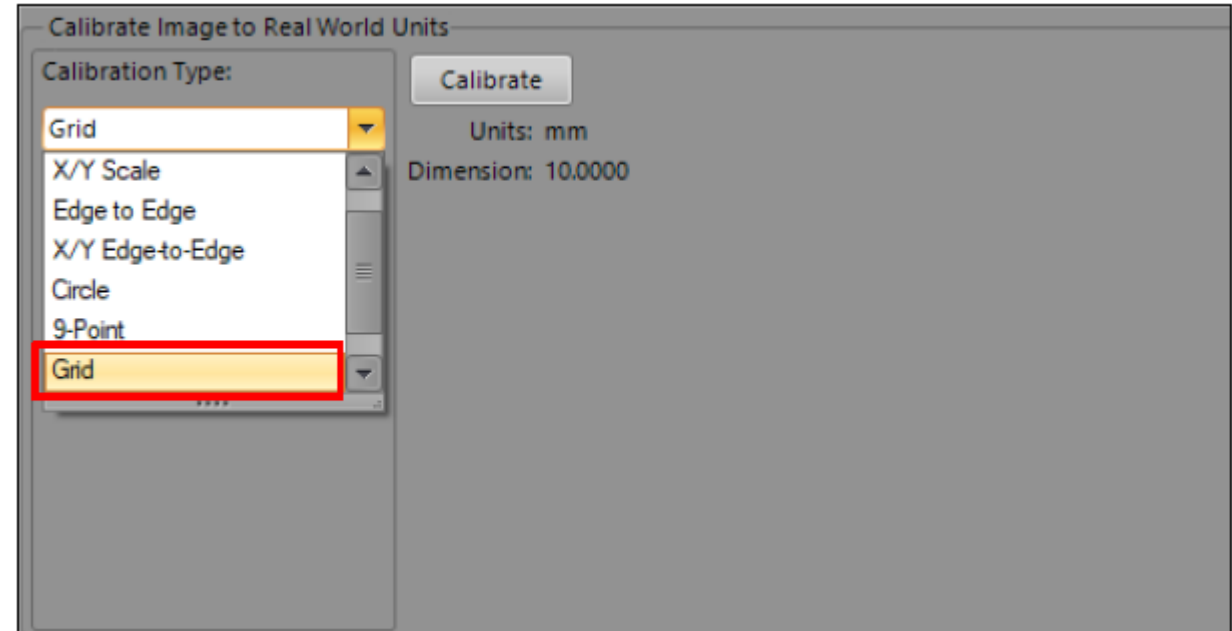
Grid Calibration

- Grid-Calibrating an In-Sight vision system enables you to more accurately locate objects at the edges of images, and provides greater mounting flexibility, including the ability to mount the In-Sight vision system up to 45° from the inspection surface.



Grid Calibration

- This is useful in machine vision applications which require a higher degree of accuracy in reporting results, such as gauging, and vision guided robotic applications. It does account for radial, barrel, or perspective / skew distortion.



Calibration Guidelines

- Keep the calibration set-up identical to the production set-up:
 - Keep calibration object and part in the same plane
 - Limit calibration to region of image features of interest
- Calibrate periodically – whenever you think that the setup may change (each shift, daily, etc.).

